T1 - Burns
Effective: February 19, 2025
Revised: July 2025

Burns may be:

- Superficial (1st Degree) red - painful (do not include in total body surface area (TBSA))
- Partial Thickness (2nd Degree) blistering
- Full Thickness (3rd Degree) painless/charred or leathery

Determine the depth and the extent (TBSA Total body surface area) of the burn. Utilize the Rule of Nines

Early intubation is required when the patient experiences significant inhalation injuries.

Circumferential burns to extremities are dangerous due to potential vascular compromise secondary to soft tissue swelling.

Burn patients are prone to hypothermia - never apply ice to cool the burn. Maintain normal body temperature and prevent shivering.

Evaluate the possibility of child abuse with children and burn injuries.

Analgesics are **not** to be administered intramuscularly.

Suspect carbon monoxide/cyanide poisoning if the patient has headache, dizziness, nausea, vomiting, decreased mental status, syncope, or chest pain or was trapped in a closed space.

Estimate TBSA. Palmer method is preferred to the Rule of Nines for estimated TBSA. The patient's palm included their fingers = 1%

For burn patients with greater than 20% TBSA, fluid resuscitation with Lactated Ringers should be initiated.

- 500 ml/hr for adults
- 250ml/hr for patients ages 6-14
- 125ml/hr for patients < 6

NOTE: In the event of arrest due to lightning strike or electrocution, follow standard ACLS protocols. Do not discontinue resuscitation in the field.

Begin Initial Medical Care and/or Multi system Trauma Care

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Remove the patient from the source and extinguish the patient if necessary.

Burns may be cooled but do not cool or moisten more than 10% of the TBSA at any one time. This may cause hypothermia. Keep the rest of the body covered to maintain body heat.

Thermal Burns:

- Remove all rings, jewelry and constricting clothing. If clothing is burned to patient do not remove.
- Monitor airway. If rapidly swelling shut and intubation is not likely, nebulize **1:1 epinephrine 0.5 ml/kg to a maximum dose of 5 mg.**
- Be prepared to perform a cricothyrotomy.
- Cover the patient with a clean sheet or burn sheet.
- Initiate IVF with Normal Saline. Do not exceed 20 ml/KG in the adult or child unless in shock.
- See Pain Management protocol.

Chemical Burns:

- Scene decontamination as needed in accordance with FFEMS policies and procedures. Remove clothing at scene.
- Flush vigorously with water or normal saline. Flush in a manner to not contaminate other surfaces.
- Alkaline burns require continuous flushing en route.

Electrical Burns:

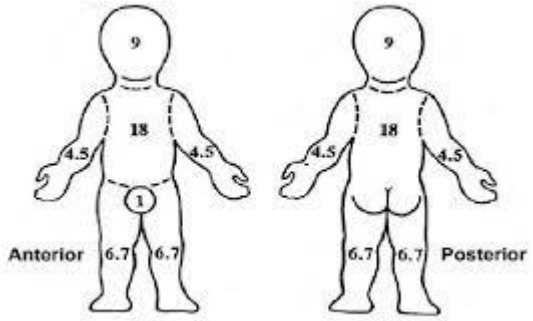
- DO NOT contact patient until you are certain the source of the electrical shock is disconnected.
- Attempt to locate contact points (generally there will be two or more.) A point where the patient contacted the source and a point(s) where the patient is grounded. Sites will generally be full thickness. **Do not refer to as entry and exit sites or wounds**.
- Cardiac Monitor: Anticipate ventricular or atrial irregularity including VT, VF, atrial fibrillation and / or heart blocks. Treat according to protocol.
- Attempt to identify the nature of the electrical source (AC / DC,) the amount of voltage and the amperage the patient may have been exposed to during the electrical shock.
- Other care as in Thermal Burns

Lightning Strike:

- Utilize Reverse Triage for multiple victims.
- Expect prolonged respiratory arrest
- Cardiac Monitor: Anticipate ventricular arrhythmias or primary asystole.
- Other care as in Thermal Burn.

Radiation Burns:

- Radiation burns with early presentation are unlikely, it is more likely this is a combination event with either thermal or chemical burn being presented as well as a radiation exposure. Where the burn is from a radiation source, it indicates the patient has been exposed to a significant source, (> 250 rem).
- Patients experiencing radiation exposure are not a risk. Cross contamination is only a threat with external and internal contamination.
- Decontaminate and transport in accordance with FFEMS policies and procedures.
- Lifesaving procedures take precedence over decontamination.



Rule of Nines

· Seldom do you find a complete isolated body part that is injured as described in the Rule of Nines.

· More likely, it will be portions of one area, portions of another, and an approximation will be needed.

· For the purpose of determining the extent of serious injury, differentiate the area with minimal or 1st degree burn from those of partial (2nd) or full (3rd) thickness burns.

· For the purpose of determining Total Body Surface Area (TBSA) of burn, include only Partial and Full Thickness burns. Report the observation of other superficial (1st degree) burns but do not include those burns in your TBSA estimate.

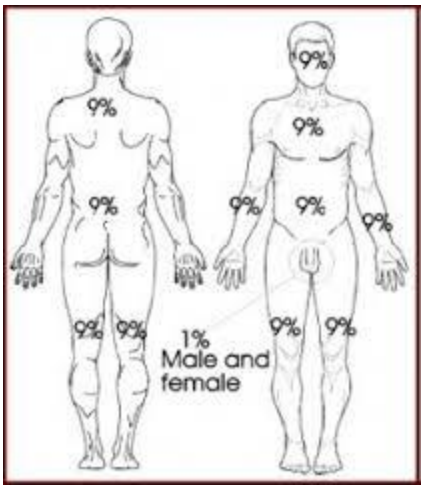
· Some texts will refer to 4th 5th and 6th degree burns. There is significant debate regarding the actual value of identifying a burn injury beyond that of the superficial, partial and full thickness burn at least at the level of emergent and primary care. For our work, all are included in Full Thickness burns.

· Other burn classifications in general include:

· 4th referring to a burn that destroys the dermis and involves muscle tissue.

· 5th referring to a burn that destroys dermis, penetrates muscle tissue, and involves tissue around the bone.

· 6th referring to a burn that destroys dermis, destroys muscle tissue, and penetrates or destroys bone tissue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